

**Smart IoT-Based Adaptable Motor Control and Safety System with
Real-Time Monitoring for Electric Vehicle Prototype**

Mini Project report submitted in partial fulfillment of the requirements for the
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IN
ELECTRICAL AND ELECTRONICS ENGINEERING**

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CERTIFICATE

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ABSTRACT

In this paper, the development of Smart IoT-based adaptive motor control and safety system will be discussed. In today's electric vehicles, having a motor control system that efficiently and reliably runs motors regardless of environmental changes is crucial for enhancing the effectiveness of the entire system. However, existing motor control systems do not possess the ability to quickly adjust to any changes in factors like load, speed, and external disturbances, hence leading to low efficiency, increased wear and tear, and even system failure. This makes it necessary to monitor all operating conditions and make adaptive adjustments as the only way of ensuring stable operation.

The design goes further to incorporate other vital aspects of operation for the benefit of efficiency and reliability. For instance, the system can detect abnormal conditions like imbalance or collisions and trigger protective measures as well as inform the user of the current status of the system at any particular time. Also, due to the incorporation of IoT into the system, remote monitoring becomes feasible for the convenience of operation. Users are therefore able to remotely monitor and control system operations at any time from anywhere.

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LIST OF ABBREVIATIONS

MCU – Microcontroller Unit

IoT – Internet of Things

Wi-Fi – Wireless Fidelity

ESP – Espressif Systems Processor

ADC – Analog to Digital Converter

DAC – Digital to Analog Converter

PWM – Pulse Width Modulation

I/O – Input/Output

DC – Direct Current

LCD – Liquid Crystal Display

LED – Light Emitting Diode

TX – Transmit

RX – Receive

EEPROM – Electrically Erasable Programmable Read Only Memory

VCC – Voltage Common Collector

GND – Ground

IDE – Integrated Development Environment

GPIO – General Purpose Input Output

SDA – Serial Data Line

SCL – Serial Clock Line

RPM – Revolutions Per Minute

NO – Normally Open

NC – Normally Close

CHAPTER - 1

1.1 INTRODUCTION

The significance of EVs is increasing with each passing day, owing to the rising need for energy-efficient and environmentally sustainable modes of transportation. Motor control in any motorized vehicle is an important factor that influences its performance, efficiency, and dependability. In practice, however, the motor experiences continually changing environmental conditions, which makes it difficult to achieve effective motor control. Traditional methods of motor control fail to deal with dynamic conditions, which results in inefficiency and loss of energy

The latest development involves incorporating motor monitoring with adaptive control measures to enhance the efficiency of the system. In one experiment, a monitoring mechanism based on IoT technology was used for a motor, whereby sensors would collect the operational data and relay them to a distant computer for analysis and management. The system improves access for the users while ensuring continuous monitoring of the motor's performance; however, it does not offer any means of adjusting the motor's operation during the process. In another experiment, researchers developed an adaptive control system for a motor, employing feedback methods to manage the motor's performance. Although the system can increase the efficiency of the motor while minimizing mechanical wear and tear, it lacks safety and remote monitoring measures.

1.2 Problem Statement:

The motor control systems in vehicles do not work well because of all the changes that happen when they are being used. Things like the load changing the speed going up and down and things outside the vehicle affecting it all make it hard for the motor control systems to work properly. The usual ways of controlling the motors are designed for when things are steady and do not change much. They cannot adapt to all the changes that happen. The systems do not work efficiently. They also respond poorly, use much energy and put too much stress on the parts, which can make them break down sooner.

Another big problem with the systems we have now is that they do not have safety features. Sometimes things can go wrong like the vehicle being out of balance or getting a shock or even being in an accident. If the system does not know how to handle these situations it can get badly damaged. Also many of the motor control systems do not let you watch what is happening in real time or control them from far away. This makes it hard for people to keep an eye on the system and fix any problems that come up.

The old systems also do not have the ability to make decisions, which means they cannot work well when things are changing. This shows that we need a motor control system that can adapt to changes has safety features and lets you watch what is happening in real time. Motor control systems, like this should be able to work efficiently work reliably and let people control them from far away. This would fix the problems we have with the systems we are using now. Motor control systems need to be improved so that motor control systems can work better..

1.3 Objectives

- The goal of this project is to create a system that can control motors in electric vehicles. This system will keep watching things like how the motor is working and make sure it keeps running smoothly even when things change. It will do this by using controls that can adjust to different situations.
- The system wants to make the motor work better and use energy when things are changing. It also wants to make sure the system is safe by finding problems like if something's not balanced or if something hits the motor. Then it will do what it needs to do to protect the system.
- This project also wants to let people check on the system and control it from away using special internet technology. This way people can see how the system is working and make changes from else. The main goal is to make motor systems work better be more reliable and safer, for users.

1.4 Proposed System

This proposed system describes an adaptable method for controlling electric vehicles motor's work. System would always watch to its parameters and adapt operation so that motor could be able to work with desired condition and maintain its stability. It would be possible to manage motor's work at variable loads and speeds since in suggested system specific methods will be used to control operation. Performance of motor can be improved, and wear of motor parts can be reduced.

In addition to that it is possible to view the running condition, find out potential faults and then the required steps may be adopted for eliminating them, if they cause damage. In addition to that the motor operation may be watched even through Internet technologies. Hence, you can observe workings of electric vehicle's motors and correct them if required. Such method is good for both electric vehicles and other kinds of motors, as system is adaptable to every conditions and it works reliably and controllable remotely. The proposed system should be beneficial to use to control electric vehicle motors.

1.5 Significance of the Study

The importance of this work is that it can make motor control systems in vehicles work better. These systems need to be efficient and safe. The old ways of controlling motors have some problems. This new approach makes the system work better when things are changing. It can use energy better. Make the motor last longer. It also has safety features to prevent damage when something goes wrong.

The system also uses the internet to let people check on it and control it from away. This makes it easier for people to use and get to. It is a system for things we use today. The ideas in this work can also be used in places like factories and other systems that need motors to work well.

1.6 Structure of the Report

This report is divided into parts. The report covers electric vehicle motor control systems.

Here is how it is organized:

This part explains the electric vehicle motor control systems project. It talks about the problems with electric vehicle motor control systems. It also says what we want to achieve with electric vehicle motor control systems and how we plan to achieve it.

Literature Review: This part looks at electric vehicle motor control systems that other people have made. It talks about motor control systems. How to control them. We also look at what's missing in their work on electric vehicle motor control systems and why we need to do something new with electric vehicle motor control systems

System Methodology: This part explains how our electric vehicle motor control systems work. It describes the plan, for electric vehicle motor control systems. It also says how we made the electric vehicle motor control systems control the motor keep it safe and connect it to the internet.

Implementation and Results: This part tells you how we built the electric vehicle motor control systems and what we found out when we tested electric vehicle motor control systems. We talk about what worked and what did not with electric vehicle motor control systems.

Future Scope: This part sums up what we did with electric vehicle motor control systems and what we learned about electric vehicle motor control systems. We also talk about what we can do to make electric vehicle motor control systems better.

CHAPTER – 2: Literature Review

The work by Krishnan and Lee in 2019 focused on making Brushless DC motors used in vehicles work better. They made a controller that used logic to handle changes in load and speed. This made the motor work smoothly and be more stable.. They did not talk about watching the motor in real time or keeping it safe which is very important for using it in real life.

Patel, Sharma and Gupta made a system in 2021 that used the internet to watch the motor and see how it was working. This system let people check on the motor from anywhere. They could see if something was wrong and fix it.. This system only watched the motor and did not make it work better when things changed.

Zhang, Liu and Chen made a system in 2022 that used feedback to make the motor work better when the load changed. This made the motor work better and last longer.. It did not let people watch the motor from far away or keep it safe.

Kumar and Reddy made a system in 2020 that used sensors to see if something was wrong with the motor. If something was wrong the system would turn off the motor to keep it safe. This showed how important it is to keep the motor safe.. It did not make the motor work better or let people watch it from far away.

Singh and Verma made a system in 2021 that used the internet to let people watch and control the motor from their phones or computers. This made it easier for people to use the motor.. It did not make the motor work better when things changed or keep it safe.

Rahman and Das made a system in 2023 that used the internet to watch the motor and make it work a little better. This was a start but it did not make the motor work much better or keep it very safe.

Mehta and Kulkarni also worked on this in 2022. They made a system that used ideas to make the motor work better. It used real time feedback to make decisions. This made the motor work better when things changed.. It did not let people watch the motor from far away or keep it safe which is important for using it in real life. Brushless DC motors, like these need to be safe and work well.

2.1 Introduction

Motor control systems are really important these days especially when it comes to vehicles. Electric vehicles need to be good at using energy work all the time and be safe for everyone. The ways we do things now are not good enough for vehicles because everything is changing so fast. Electric vehicles require good motor control systems to be efficient and dependable. Motor control systems, for vehicles have to be safe too. So people are looking into connected systems.

With the Internet of Things and new control strategies researchers have come up with ideas to make motor systems better and safer. This chapter is about the things related to using the Internet of Things to monitor motor control systems, measure motor speed, detect overloads and protect the systems. It gives a picture of what people are doing now and what the problems are, with these approaches. Motor control systems are the focus here and understanding motor control systems is crucial. Motor control systems and their development are still being researched.

2.2 IOT based motor monitoring systems

The use of IoT-based motor monitoring solutions is applied in order to keep an eye on the motor settings in real time. These monitoring systems incorporate sensors used to gather information regarding issues like performance, temperature, and load of the motor. After gathering this information, it is relayed through the internet. It becomes easy for one to be able to know about the current status of the motor and hence be able to spot any malfunctions that occur..

Numerous studies have been conducted that have shown that IoT is effective in remotely monitoring and controlling devices through computers and phones. However, most of these solutions are efficient in data collection and display but they lack the ability to change anything by themselves. Therefore, while we can have better control over the processes occurring, it

might not be possible to handle any change that occurs during the process. It is important for IoT-based motor monitoring systems to be adaptive to change.

2.3 Motor Speed Measurement Techniques

Measuring the speed of a motor is really important for the motor to work well and run properly. The motor needs to be checked. There are a ways to measure the speed of the motor. You can use tools like tachometers, optical encoders and Hall-effect sensors, which are extremely useful for checking how fast the motor spins; something that is crucial if the motor runs smoothly. So this gives us some idea what is actually going on at the far end of what is to ensure the motor is being properly controlled.

The encoder based methods are extremely accurate, on the other hand the sensors are extremely cheaper for a real life applications. However some of these methods do not work well when there is a lot of noise around or if things are not lined up correctly and Also measuring the speed of a motor and using that information to control it in a way is not always done, which means these systems do not always work well in situations that are changing. Motor speed measurement is important. We need to make sure we are doing it in a way that works well with the control systems.

2.4 Load cells based overload detection

Over load detection is very important for the safety of the system and avoids damage to the mechanism. Above load cells have been used to sense the force being exerted onto the system and to get a reliable value for overload detection. These sensors can transform the mechanical forces applied on them to electric signals which can be used for over load condition detecting. The load cell based system have been proved to accurately detect over load condition and provide the required information for take precautionary action. The systems only detected the overload; they don't contain adaptable control systems which respond to the overload.

2.5 Motor Protection Systems

Motor protection system is installed in order to keep the whole system intact and detects abnormal conditions and failure such as overheat, overload, short circuit, mechanical breakage etc, and responds to the system. There are many systems of protection which are utilized in order for adequate protection to be achieved. protection is accomplished via thermal protection, fault limitation and detection.

The new system developed above also include the tilt detection and collision detection function, it may contribute to the safety of mobile application, like the electric vehicle. However, while this kind of protection against the damage of the motor itself is valuable, it is implemented separately from the control and monitoring system so it loses some effectiveness.

2.6 Research Gap

Through the literatures above, we know that the systems are used in a function which is,. Observation, speed determination, excessive speed safety device. On the other hand IoT based system concerned monitoring, but did not control the system adaptive according to the conditions whereas control systems control the system perfectly, but is lack of safety or network connection while protection system is used to operate with no connected system. It shows the importance of an integrated system where adaptive motor control, real-time monitoring, over-load detection and safety can be coupled to provide better and efficient interaction.

2.7 Conclusion

In this chapter, several aspects of monitoring, control and safety of motor systems have been discussed. Though great progress has been made in each individual aspects, the biggest drawback is the lack of integration among these three. The research gap found hence shows that, an integrated motor monitoring control and protection system is required. This leads to the proposed work given in the subsequent chapters.

CHAPTER - 3 System Design and Methodology

3.1 Overview

This proposed system can be characterized as a smart, reliable and flexible system of motor control for the electric vehicle, designed for high reliability. By merging these various aspects (sensing, control, safety and communication) in a self contained unit, the motor system is operated appropriately for the particular application. In a practical scenario, these parameters like the load, speed, and external disturbances affecting the performance of motor control, are continuously varying. The system has to adapt with them by observing parameters and controlling the motor. The whole sequence diagram shows how data acquisition, process, control, and security mechanism is handled under normal and abnormal conditions as shown in Fig 3.1.

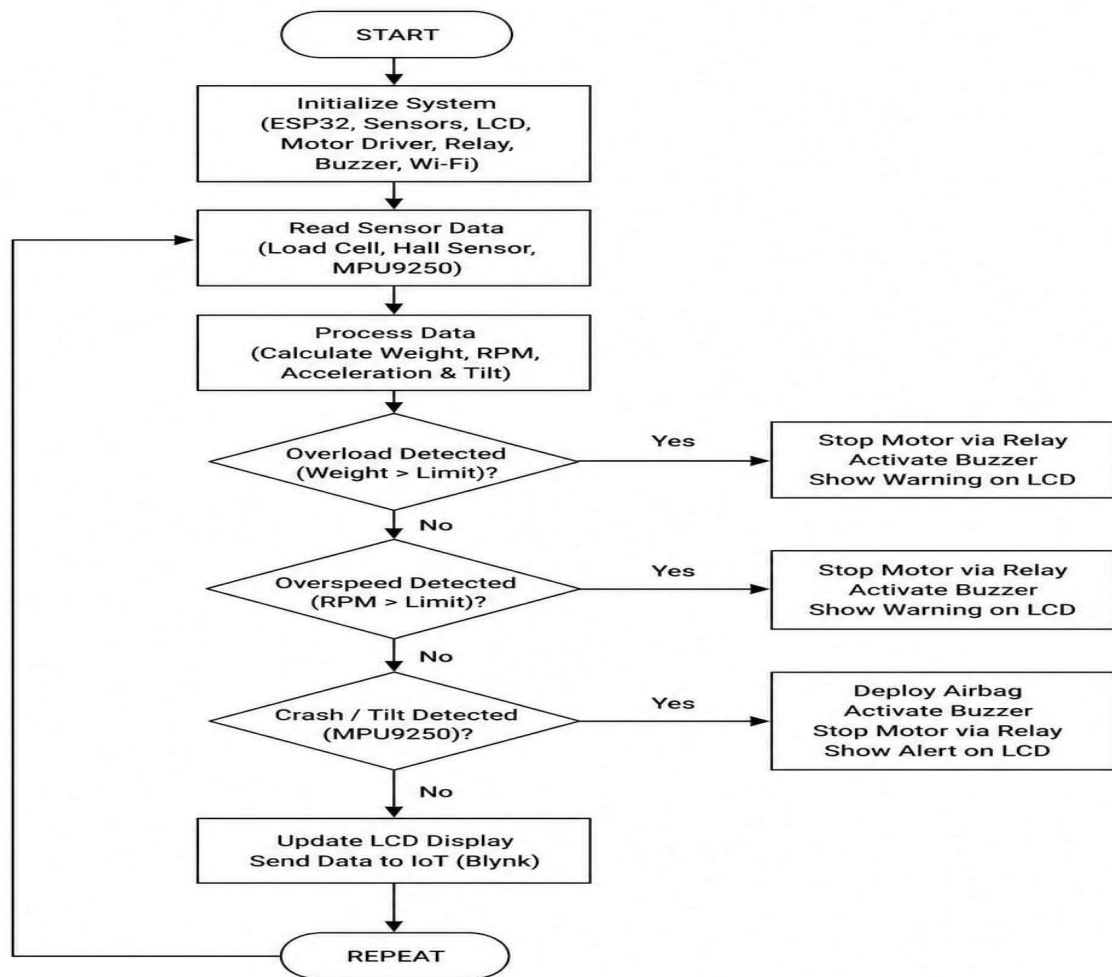


Fig-3.1 Flow Chart

3.2 System Architecture

System architecture is designed with multiple layer of sensing, processing, control and communication modules that are used to ensure its high performance and reliability. Sensing layer which covers different type of sensor used for collecting the instantaneous readings of the load, speed and system attitude. Information will send to the processing layer for the control system to analyzes it and decide what to do with corresponding to the logic and adaptive control approach.

The control layer regulates motor movement via appropriate control signals which is generated by the control unit. The signals is sent to the motor driver that acts as an interface between the control unit and the motor. The speed and working of the motor can be effectively regulated by control signals in to it. At the same time system also continuously monitor for the safety issues. The undesirable condition that occurs would directly trigger corresponding protection mechanisms that turn off the motor and issues warning.

Communication layer that integrated with IoT capability allowing the data to be transfer to remote station for monitoring and control. This would be able to show and control the instantaneous information of system using remote device. All the modules works smoothly and effectively with one another, and it leads to a safe and efficient system performance in different dynamic condition

3.3 Hardware Description

3.3.1 ESP32 Microcontroller

The main controller and interface of the project will be the ESP32. It's a SoC developed by Espressif Systems that was chosen to meet the system's needs for two processor cores and a true multi-tasking architecture. Both of the processor cores will be utilized for the stable functioning of the system; one will be dedicated to handling the high-frequency/high-speed motor control loops and safety interrupts, while the second will run the wi-fi stack independently to achieve real-time IoT data logging and cloud sync.

Core Architecture and Processing Power

The core of the ESP32 chip is a Xtensa Dual-Core 32-bit LX6 microprocessor which can be clock set between 80Mhz and 240Mhz. The clock speed can be kept high in this system, to keep any latency from the load cell sensor readings low, as the high clock speeds and 520kb on board SRAM means that readings can be buffered, waiting until there is a wi-fi connection, meaning no safety telemetry can be missed.

Peripheral Interfacing and Connectivity

The ESP32 microcontroller, is quite different from a basic 8bit microcontroller; it is integrated with an excess of high-speed peripheral interfaces. The peripheral interfaces used for this system were;

- * PWM - For an extremely high precision speed control with the motor via the L298N.
- * ADC - For the high output signal from the amplified load cell signals.
- * I2C - The two wire serial interface, that was chosen in order to have only the use of two pins for the LCD display.

Below in Fig 3.3.1, shows the internal and the pin layout of the ESP32 development board, so as to enable one to observe, what pins have been used and what routes will be used in the system.

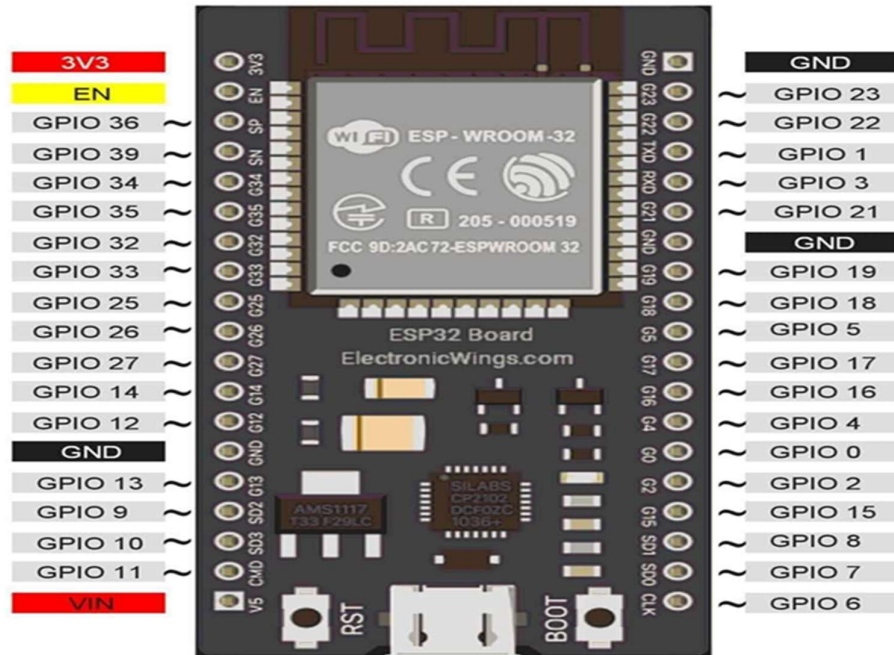


Fig 3.3.1: ESP32 Development Board Pinout and Internal Architecture

Power Management and Efficiency

The ESP32 has various power modes that could be utilized within the system to conserve power while the motor system is idle (Clock gating and deep sleep, and light sleep modes). The logic level of the system is 3.3V, but the development board has an onboard LDO which enables the system to run from a 5V power source.

Reasons for Selecting ESP32

Selected ESP32 rather than Arduino Uno due to;

- * On-board wireless communication (don't have to buy a separate shield).
- * Higher resolution (12-bit ADC).

3.3.2 HX711 Load Cell Amplifier

The HX711 is a sensitive tool used for converting very small signals into a digital form. It is most commonly used in systems where it is used to measure weight. The HX711 connects to a load cell in order to measure force or weight. A load cell generates a small signal, but this signal is often too small for a microcontroller to read. For this reason, the HX711 is used to amplify and convert this small signal into useful data.

The HX711 can easily interface with controllers like the ESP32 and the Arduino because it has only 2 connections needed: data and clock. Because it is very simple to use and also very sensitive, the HX711 can be found in machines that detect the weight of a load in systems that do process monitoring in a factory, and systems that automate different things.

In this project, the HX711 is used to read the signal coming from a load cell used to detect the amount of force being placed upon the motor. The load cell gives off a signal, that is processed by the ESP32. When the ESP32 detects that there is too much force, it automatically turns on the safety system. The load cell works with the HX711 to allow the ESP32 to know how much force is on the motor. The module used for this project is displayed in figure 3.3.2.

In this project load cell is used to continuously determine the load on the motor. Data from load cell is then transmitted to the ESP32 through HX711 module. If load exerted on the motor is greater than the safe limit, the system would be able to determine the load is exceeding its limit and appropriate action to take in order to avoid damage to the motor will be executed. The load cell helps to monitor the load being exerted on the motor. Load cell helps the whole system to function working properly. The Loadcell used in this system is shown in Figure 3.3.3

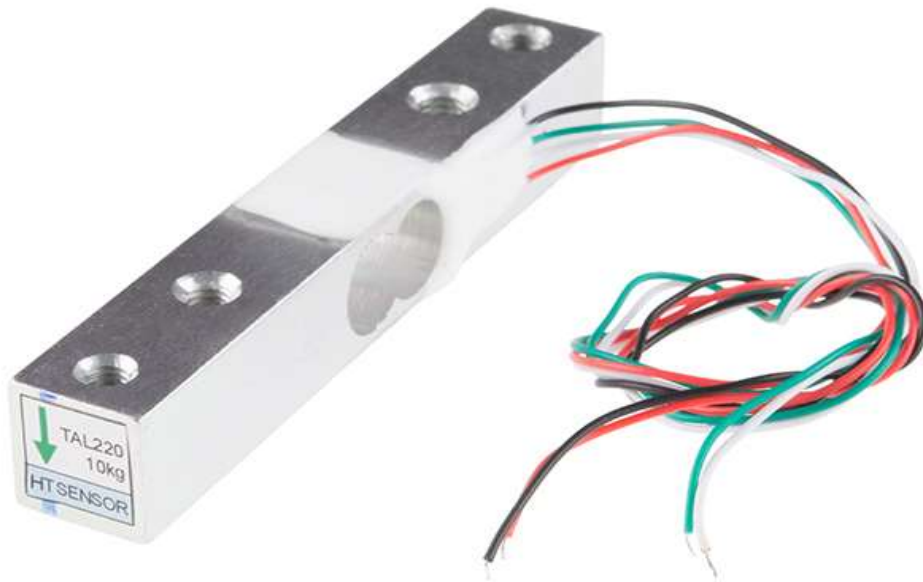


Fig 3.3.3 LOAD CELL

3.3.4 L298N Motor Driver

The L298N is a motor driver module, this module is used to control DC motor. It is the device connecting between the microcontroller and the motor. The ESP32 do not has the capacity to directly drive a motor hence the use of motor driver. It accept the low power signals that the controller send. It will convert the low power signal into high power signals that the motor is capable to handle.

The module can control up to two DC motor and one stepper motor at one time. It can move the motors forward, backward, stop, control the speed with a PWM signal. Vary the speed of motor by varying the duty cycle of the PWM signal can make the motor runs slower and faster. The module can vary the speed with this ability to reduce wastage of energy and to increase efficiency of working.

In this project, the L298N motor driver helps to control a DC motor according to the signal that the controller send. The speed of the DC motor can vary based on the situation e.g. How hard is the motor is working. With this ability, the system is able to operate efficiently. Also, it can protect the motor and the system if there is an issue with it e.g. Too hard to work for the motor, and the controller has to stop the motor. The L298N motor driver module that is use for the project is shown as in the figure 3.3.4.

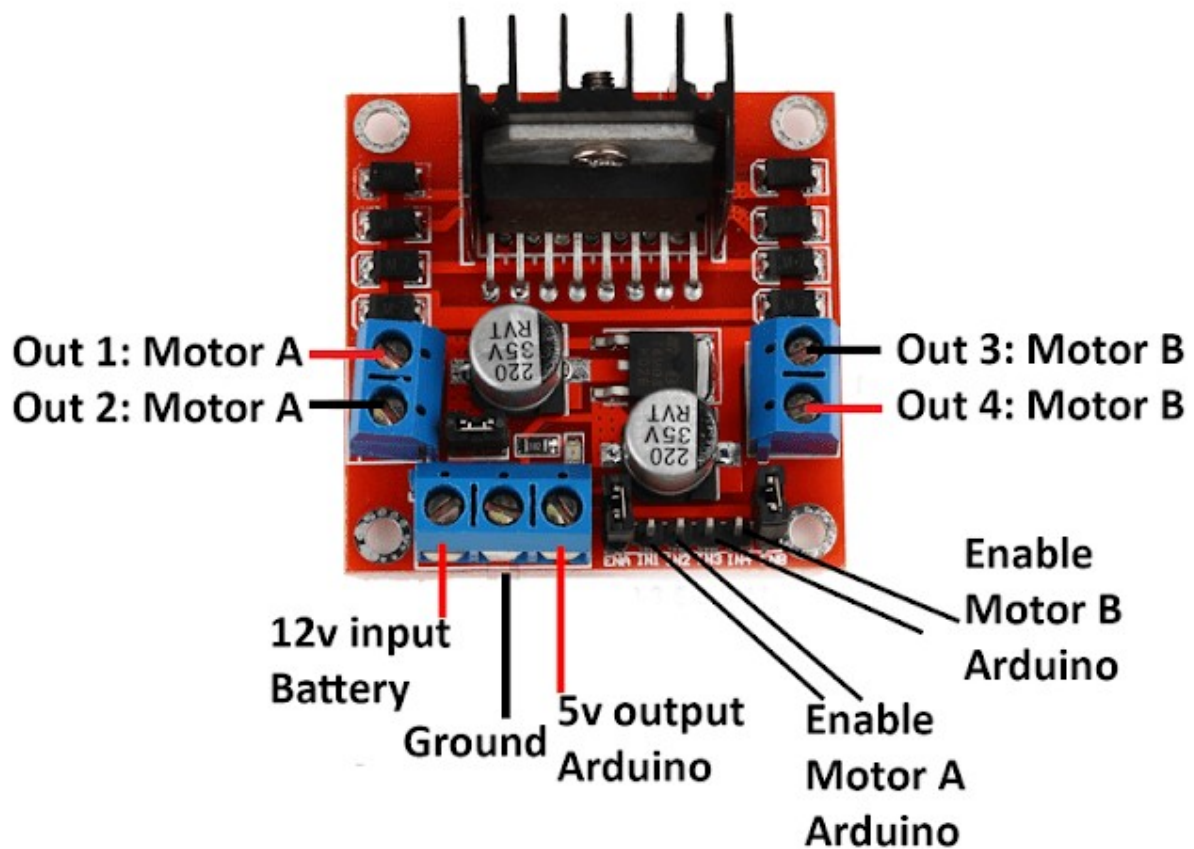


FIG 3.3.4A L298 MOTOR DRIVER

L298N Motor Driver Circuit Description

The L298N motor driver module is a useful part for controlling DC motors. In Figure 3.3.4B you can see how the different parts within the L298N motor driver connect up to one another. This part is useful as it can control the speed and direction of the motor. It can be connected up to and receive signals from the ESP32 microcontroller. The L298N motor driver consists of two H-Bridges. It is these circuits which allow the motor to spin in both forward and backward direction. There are pins on the L298N motor driver which control the direction of the motor; they are IN1, IN2, IN3 and IN4. When signals in a variety of combinations are sent to these pins on the L298N motor driver then the motor can change direction. The L298N motor driver also has enable pins called ENA and ENB. These pins on the L298N motor driver control the speed of the motor. They get PWM signals from the microcontroller like the ESP32. This helps the motor speed up and slow down smoothly. When the enable pins on the L298N motor driver are HIGH the motor is on. When they are LOW the motor stops.

The power supply part of the L298N motor driver circuit has VCC, ground and a voltage regulator like the 7805. This makes sure the power supply to the L298N motor driver and control circuit is stable. There are also capacitors in the L298N motor driver circuit. They filter out noise. Keep the voltage levels stable.

There are diodes across the motor outputs on the L298N motor driver. They protect the motor from damage. They stop the EMF from hurting the motor when it stops or changes direction.

The output terminals like OUT1 and OUT2 are connected to the motor. The motor works based on the control signals it gets from the motor driver.

The L298N motor driver circuit is really helpful in this project. It controls the motor efficiently by adjusting the speed and direction of the motor. It makes sure the motor operates smoothly. The detailed circuit of the L298N motor driver is shown in Figure 3.3.4B

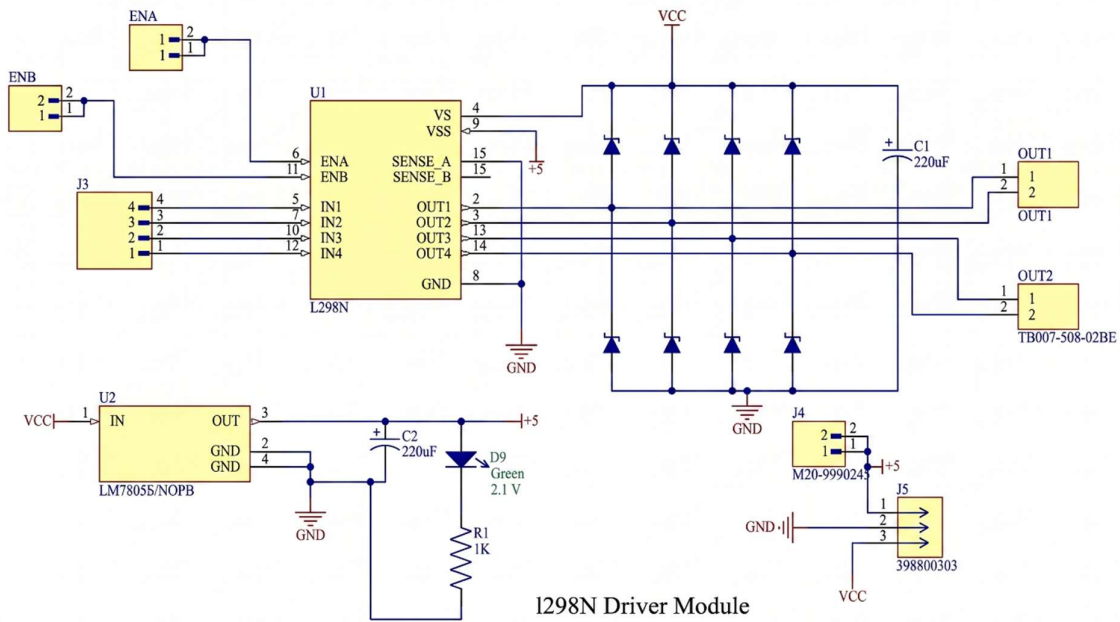


FIG 3.3.4B L298 MOTOR DRIVER

3.3.5 LCD Display (20×4)

The 204 LCD is display capable of displaying twenty characters each over four lines. Used to display what can be easily read from a computer's display (e.g. Current readings of a sensor, error messages). This 20*4 LCD module easily connects up to a ESP32, Arduino, or even another system.. Various methods can be used to connect to the system. It is generally more convenient to use the I2C connection method as it requires fewer wires and fewer connections in the entire system. The 204 LCD plays a critical role in an system as it displays real time information to the user. Such information include the load on the system, speed of motor, etc.,

it is constantly being updated and if something goes wrong (overloads the system) it will display an alert on the 204 LCD. The 204 LCD is a better option as it can display many things on the screen at the same time thus simplifying it for the user to understand and view. Figure 2.6 displays the 204 LCD that was used in the project.

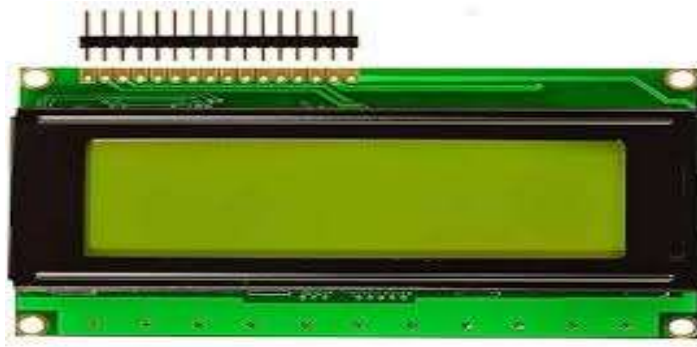


FIG 3.3.5 LCD DISPLAY

3.3.6 Relay Module

Basically, the relay module works as a switch, it allows a signal from the microcontroller to turn a high power load such as a motor on/off. When you are connecting the microcontroller to the high power device there must be some type of switch in between; these two parts form the control circuit and the load circuit. This protects sensitive components like the ESP32.

A relay module is made up of a couple of significant components. It obviously contains the relay switch. But it will also have a transistor, to amplify the signals. It also contains a diode that stops potential damage to the circuit due to voltage spikes. It has input pins that are able to read signals and output pins which can be used to turn the loads on and off. It also has indicators on it, which illuminate.

The relay module in our system works like a safety shield. It can be used to control the power to the motor. If all the system parameters are correct then it allows the motor to work properly, else the ESP32 orders it to switch OFF and in this way the system becomes safe.

Thus the relay module is very important in order to protect the system. The relay module which is used

in our system and the diagram of which is shown in Figure 3.3.6 is:

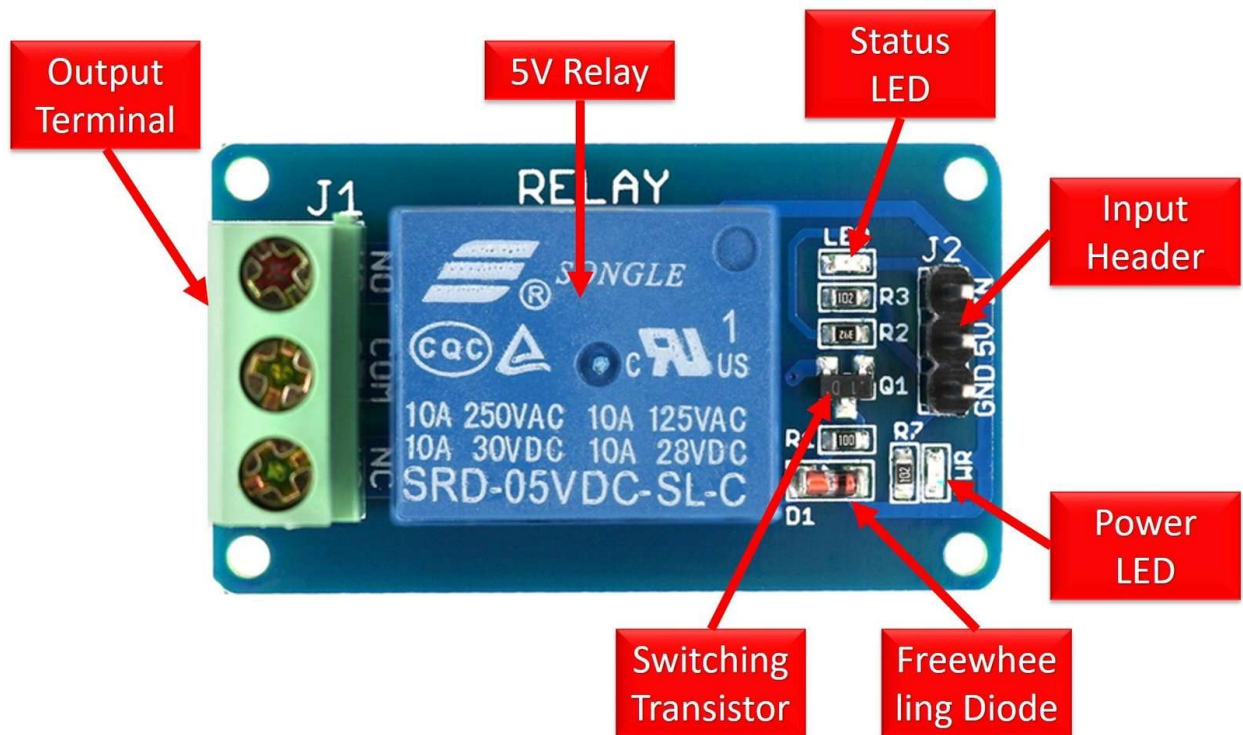


FIG 3.3.6 RELAY MODULE

3.3.7 DC Motor

The DC motor takes an electrical energy source and converts it to motion, which is rotational or circular. The motor is commonly found in applications like fans, pumps, and machinery that is employed for transportation purposes. In robotic as well as industrial automation equipment like a smart plant it has been found very handy as simple and easily controllable DC motors.

You'll notice that a DC motor can be speed controlled easily by variation of voltage which is fed to the system. One method to vary speed can be using technique such as PWM (Pulse-Width Modulation), the speed may be varied according to needs, the DC motor can rotate clockwise as well as anti-clockwise which can be controlled using circuit based systems such as an L298N module which switches the direction from forward and backwards accordingly. In the presented project the work is This is done by DC motor by itself. The system is supposed to check if its working at normal speed. The sensors that check the speed of the DC motor then supply that information to the brain of the system such as ESP32, which then check the readings

of the sensors supplied by it and sends the command to the motor. This ESP32 is the "brain" for the system and controls the DC motor. If something goes wrong with it e.g. Working hard then slows down or stop to function properly. The DC motor that we have used is displayed below at Figure 3.3.7.



FIG 3.3.7 DC MOTOR

3.3.8 MPU6500 Sensor

MPU6500 is a motion sensor that consists of 3-axis accelerometer and 3-axis gyroscope. The sensor can detect movements and orientation along the x,y, and z axes and monitor deviations in position, tilt and sharp acceleration. The MPU6500 is applied to detect unexpected accelerations or movements and based on such change, it can detect collision and execute the safety actions such as airbag deployment. Figure 2.9 depicts the MPU6500 sensor utilized in this project.



FIG 2.9 MPU6500

3.4 Circuit Diagram

This section explains the total connections among all modules present in the system, which consist of sensors, control unit, motor driver and output modules. It details how the sensing modules are interfaced with the control module and how control signals are fed to the motor through the driver circuit. Also, display, buzzer and relay module connections for user interface and safety control are shown.

The correctly designed circuit allows stable system operation and proper communication between the modules. The power supply section gives the necessary levels of voltage for all the modules and the signal connection allows transfer of proper values and control. The final circuit diagram of the system is shown in Fig 3.4

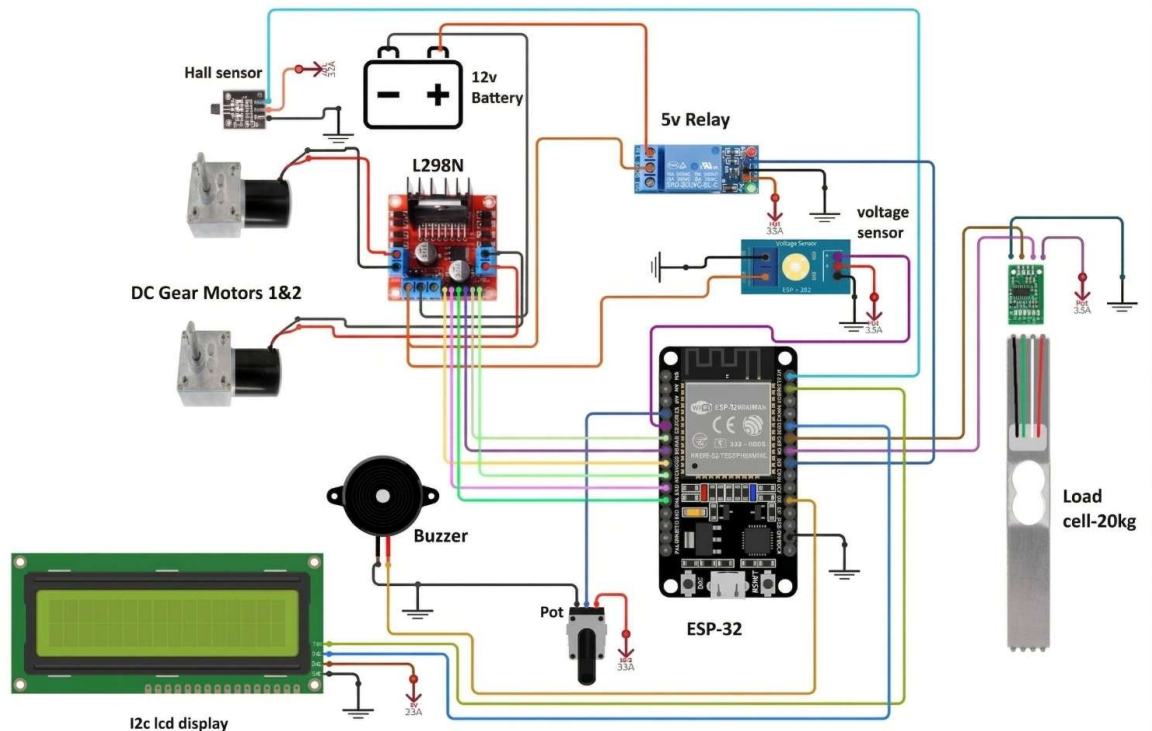


Fig 3.4 Circuit Diagram

3.5 Working Principle

The working of the suggested system is based on continuous sensing of motor operational characteristics and using this information for adaptive control of the motor speed in real time. When the system is powered on, each of the sensing and controlling parts of the system will be switched on. The control unit starts collecting information from the sensing units.

The loading conditions of the motor are provided by a loading sensor, the speed by a speed sensor, and further inputs are provided by some other sensing units (e.g. To know the system orientation or any external disturbances to the system).

The gathered information from each sensor are continuously given to the control unit where the analysis of this information is made and accordingly a specific operating condition of the motor is deduced.

The control unit generates suitable control signals and tries to maintain the system in operating conditions as far as possible, i.e. The control unit tries to maintain a specific motor speed even when the loading and speed conditions change.

Also, the system keeps checking the status of overload or system inclination and any sudden external shock applied on the system. When any undesirable situation is found, safety units such as control unit for shutting down the motor, Buzzer alarm for alarming, updating the status are used for that purpose.

In this manner the system is very effective and has a high degree of security against failure. At the same time, the system provides online update of system status (speed, load and other system conditions) at some distant location with the help of the internet and IoT technologies. It is possible to provide the commands to the system from some distant location if there is a requirement .

3.6 Software Design

The software design of the system is implemented so that data is processed effectively, decisions are controlled, and communication occurs. The system program is continually reading in data from the sensors and using the data in real time in order to figure out the state of the system. Based on the analyzed data, appropriate control signals are generated to regulate motor operation.

The software also includes logic for detecting abnormal conditions and triggering safety responses such as stopping the motor and activating alerts. In addition, communication protocols are implemented to enable data transmission for remote monitoring. The software is structured in a modular manner to ensure smooth coordination between sensing, control, and communication functions.

3.7 System Workflow

System working flow is the sequence of process while the system works. Firstly, the system gets the sensor information and sends the sensor information to control center to process. The control center processes the sensor information and decided whether the system works normally or not.

If the working condition is normal, the system is still running the control for motor and update the system state. If the working condition is abnormal, the system performs some safety process, such as stop motor and display alarm information for user. Meanwhile, the system information is sent to the remote center to supervise.

3.8 Conclusion

This chapter outlined the proposed motor control system design and methodology including the system architecture, the components of the system, the system operation and the method for implementation. With the incorporation of sensing, control, safety and IoT features, it guaranteed the effectiveness, reliability and safety of the motor control system. The system is well designed and ready for implementation and evaluation.

CHAPTER - 4: Results and Discussion

4.1 overview

This chapter describes the implementation and performance of the designed smart motor control and safety system. The system design is implemented by combining the hardware components and software functions in order to obtain the ability to achieve real-time monitoring, control and safety functions. In the design and implementation, careful coordination between the detection module, the process module, the control module and the communication module is designed.

Under the varying conditions such as speed, load variations and etc, the proposed system is operated to regulate the motor performance efficiently. Beside the motor control, the safety functions are also embedded in the system, to detect the abnormal conditions and take the appropriate protective action.

With the IoT function embedded, real-time monitoring of the system state and remote access by using an Internet connection is provided for user. At the same time, performance test in different cases are discussed to justify the efficiency of the proposed smart motor control and safety system.

4.2 Hardware Implimentation

The implementation of the system is done on hardware. The system consists of sensors, control unit, motor driver and output devices. The values of input signals such as load, speed and system parameters are fed from sensors to the control unit. The control unit then processes these input signals and generate the control signals to control the speed and operation of motor. The motor driver is used to interface control unit to the motor for the control of its speed and direction. The display unit, buzzer unit, and relay module are interfaced for user interaction and to meet safety requirements.

The wiring of the hardware has been done such a way that all devices cooperate together in order to achieve the continuous monitoring and controlling action. It is also kept ensured to provide appropriate power supply to each system as well as the earthing is also properly maintained. This sort of arrangement of devices also enables to test and repair the system easily.

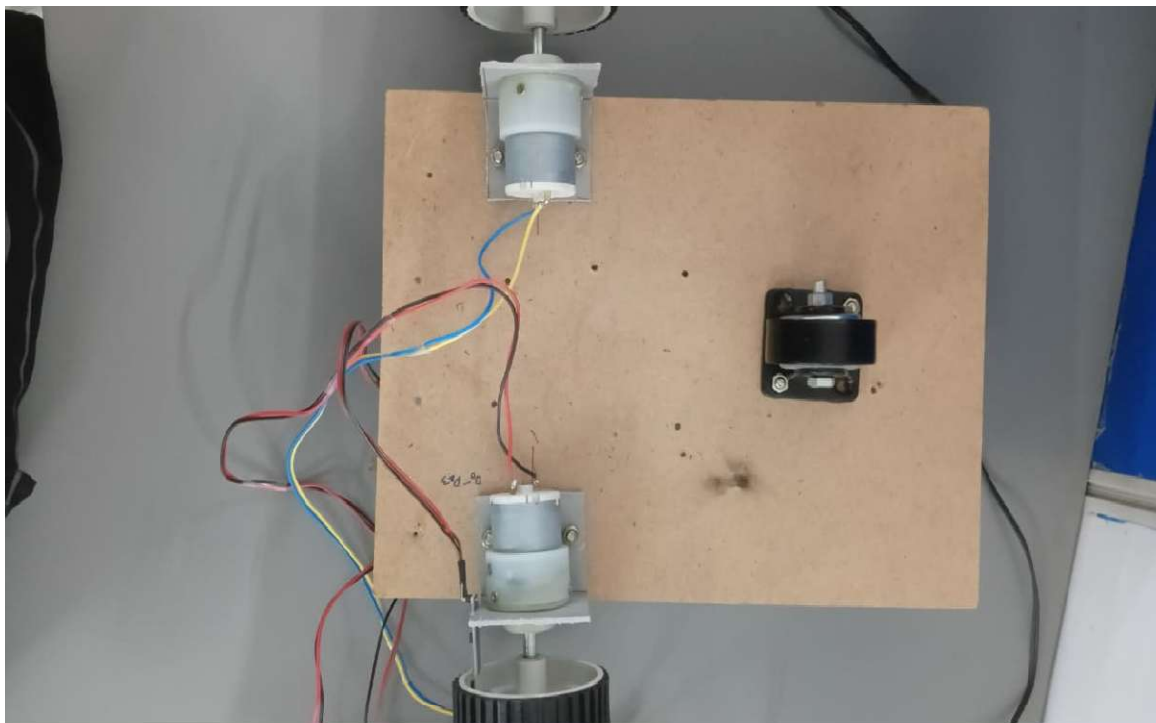
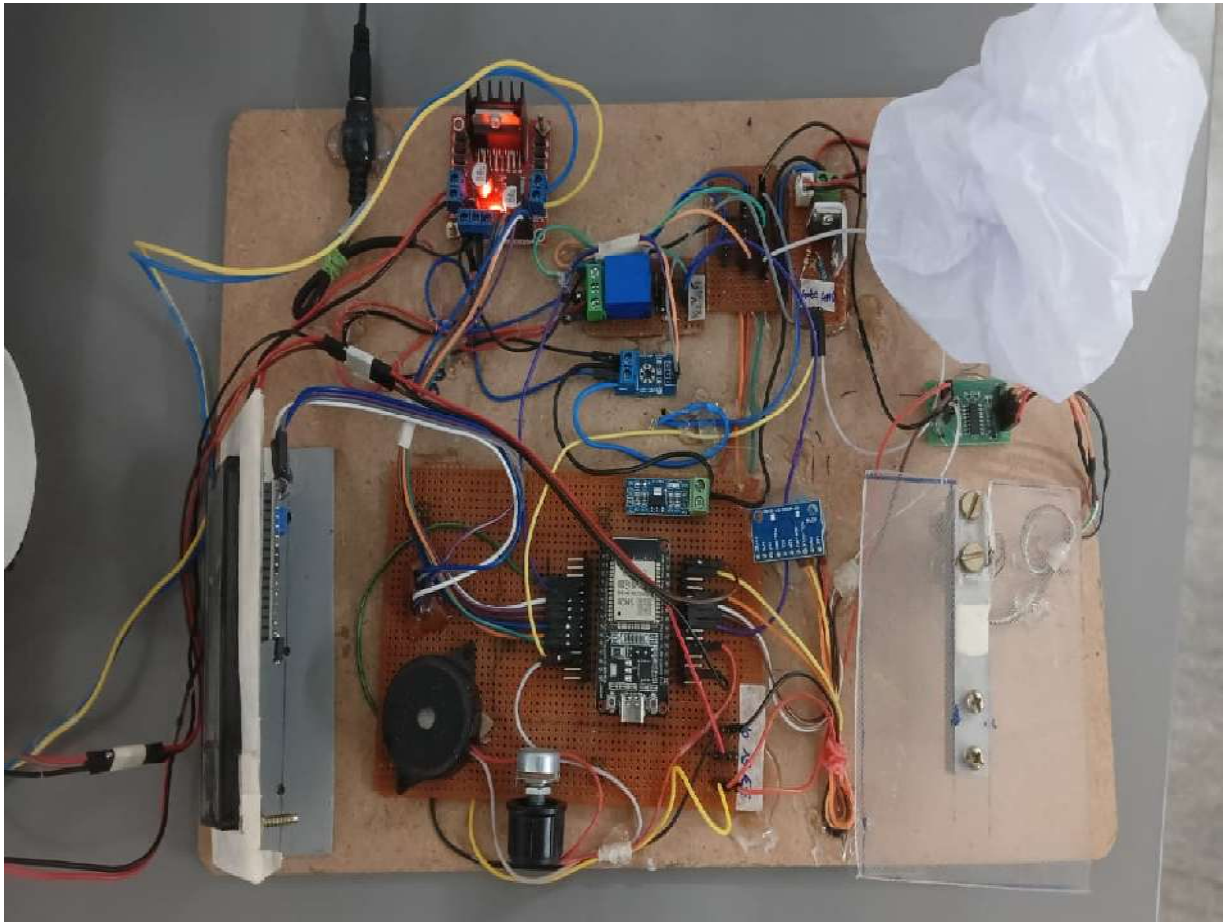


Fig 4.2 Hardware Setup

4.3 System Testing

The system was subjected to a variety of conditions so that the performance, reliability and response time of the system could be verified. To start off, the system was turned on under a normal operating condition to test that each component including sensors, the control unit, the motor driver, and the output devices were functioning correctly. The sensor outputs were observed for data acquisition and effective communication with the control unit.

After the normal operating conditions were successfully tested, the load was varied to test the adaptive capability of the system. The system recognized that the load changed and modified the output to compensate. The speed response time of the motor was measured and confirmed stable operation under varying load.

The system was then subjected to abnormality conditions. An overload was placed on the motor so that the safety features of the system were activated and monitored. The abnormal condition was recognized, the motor was stopped, and an alarm was activated. The disturbance was also tested and an abnormal condition was created in order to determine the reliability of the safety controls.

Lastly, the IoT functionality was tested by viewing the output from a remote interface. The parameters of the system could be monitored in real time and the system was able to respond to external inputs sent from the remote interface.

The system was found to function reliably under various circumstances and successfully satisfied all desired requirements for the project

4.4 Results Analysis

Test result which is as above had showed that developed system is very reliable and also work on different load conditions. Motor responded good with load change. Hence we can verify that control logic can react to real time situation, and does its job correctly. Satisfactory working speed as well; output delay time of controller upon receipt of the error signal is minimized as far as possible. As demonstrated, the motor turns freely when its control signals are based upon the inputs from the sensors.

In safety testing, system was able to detect faulty and abnormal situations such as disturbances and overloads. The system response to detect these situations were to shut off the motor and switch on the warning indications and hence system can be protected. This is because the safety circuits embedded are responsive and functional.

The IoT-based monitoring part of system was also functioned as expected to transmit data regarding motor state such as current and voltage to monitor on remote device with live updates which will bring ease and safety.

4.5 Discussion

From the above stated results, we can say that the system we proposed successfully achieved the motor performances enhancement and also enhanced the safety and monitoring capabilities. The adaptive control ensured stable and continuous operation of system in different operating condition. The safety mechanism ensures security when system faced abnormal condition. IoT allows the user to control the system from far and monitor it continuously, it increases user comfort ability. However, in order to increase the accuracy of the system and also enhance scalability, there are further improvements needed.

Chapter -5 Conclusion and Future Scope

5.1 Conclusion

As can be observed from the implementation results that the motor control system with real time monitoring can greatly improve the performance. The system can be in stable state and highly efficient with adapt control system in different condition of load and speed. It follows the principle of the conservation of energy and the stress of component is reduce so that it can increase the reliability and make control and monitor by remote. It can be very convenient to user and guarantee the safety in some situation, the system can maintain a credible performance for the interactive system under control by adaptive control system and security with the help of IoT.

The system implementation has shown that the motor control with real time monitoring provides high performance. It can follow the varying load and speed under the adapt control method in real time, so that the performance is steady and the system has a high efficiency to drive the motor. Thus the energy is conserved and the mechanical stress of component will be reduced which can guarantee the reliability. During the test of varying loads, system can get a much better result which is so important for the application such as electrical car, etc.

Safety function of the system is also be considered. By using variety sensors which attach to the system, abnormal condition of the motor working such as overloading, imbalanced working state, or the disturbance state can be sensed and suitable method can be executed, such as stopping the motor and alarm. Which can achieve the security for the entire system application.

IoT feature of the system make it be a very flexible system. The users can monitor and control it with the help of network for observing the condition of the motor working such as speed, load, working state, etc., and make decision for it in the remote, which make the work of the system be very convenient and achieve security in some specific condition of working.

5.2 Future Scope

Still, even though the proposed solution looks feasible and is the best among those that exist currently, there is no limit to the potential areas for improvement in the field. The use of advanced control methodologies, for example, fuzzy logic control or neural networks control might bring more advanced adaptive control capabilities to the project, which can be quite beneficial in a dynamic environment like that.

Use of high-precision sensors and an advanced calibration approach can provide for increased precision in data collection and analysis as well as in actual implementation of control algorithms. Another area for improvement is hardware architecture and design; smaller, less energy-consuming and easier-to-manufacture devices can be used to create a scalable solution to the problem.

There are also possibilities for improving performance using the Internet of Things and cloud computing; a cloud platform can be used to store collected data and analyze them to provide advanced decision-making capability and long-term performance evaluation, while an app interface can simplify and streamline the process of working with the system.

Moreover, with appropriate modification, this system can be used to predict system malfunctions and perform predictive maintenance, thereby increasing system reliability and reducing maintenance costs.

The scope of applications for this technology is not limited to electric vehicle propulsion systems. The same methodology can be implemented into various fields such as automation, robotics, and infrastructure in general. After some modification, this system can be used to control much higher power systems and systems of complicated architecture.

In conclusion, a smart motor control system is potentially feasible. After necessary changes are made to control algorithms, sensor mechanisms and data exchange infrastructure, this project has a great potential to prove itself useful and successful.

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